

POCONO AI, LLC

Governance-native AI infrastructure for regulated work

The AI Near-Miss Doctrine

Incident Response, Root-Cause Review, and Corrective Action for
Regulated AI Workflows

A flagship Pocono AI governance white paper for healthcare practices, law firms, compliance leaders, investors, patient advocates, and client advocates.

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Core thesis: Responsible AI is not AI that never makes mistakes. Responsible AI is AI that makes mistakes detectable, reviewable, correctable, and defensible before they become harm.

Important Notice

This white paper is an educational and governance-oriented research document. It is not legal advice, medical advice, clinical guidance, regulatory advice, or a substitute for counsel, licensed clinical judgment, professional malpractice guidance, insurer guidance, or regulator-specific compliance review. Healthcare organizations and law firms should consult their own attorneys, compliance professionals, clinical leadership, privacy officers, insurers, and professional boards before adopting any AI governance program.

Sentinel Node is discussed as an implementation example of local, evidence-preserving AI infrastructure. The doctrine described here is broader than any one product. The central point is that regulated AI use should create evidence as it works: evidence of data custody, source retrieval, drafting, review, correction, approval, export, and post-event learning.

How to read this document

- Executives should read the Executive Summary, Sections 1-4, and the Buyer Checklist.
- Physicians, nurses, attorneys, paralegals, and practice administrators should read the severity model and the incident lifecycle.
- Compliance, IT, and operations leaders should read the root-cause categories, corrective action model, and implementation roadmap.
- Investors should read the market thesis and the defensibility sections.

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Executive Summary

Generative AI is entering regulated professional work before many organizations have built the governance structures required to manage its failures. Healthcare practices are considering AI for documentation, prior authorization, denials management, chart summarization, patient communications, internal policy retrieval, and administrative overload. Law firms are considering AI for legal research, document review, contract analysis, discovery support, drafting, and internal knowledge retrieval. In both sectors, the same operational risk is emerging: AI systems can produce fluent, persuasive, incomplete, unsupported, or false outputs that appear usable before they have been verified.

The central problem is not merely that AI can be wrong. Professionals already work in environments where error is possible. The deeper problem is that many AI deployments do not preserve enough evidence to determine what went wrong, who reviewed the output, what sources were used, whether the output was grounded, whether a human approved it, whether the output left the organization, and whether the organization learned from the event.

This white paper proposes a practical doctrine for regulated AI governance: the AI near-miss doctrine. A near miss is an unsafe event that did not become harm because it was caught, blocked, corrected, or contained in time. In medicine, near misses are essential to patient-safety learning. In law, near misses can reveal weaknesses in supervision, verification, confidentiality, and candor before those weaknesses become sanctions, malpractice exposure, client harm, or court-facing embarrassment.

The doctrine advanced here is simple: responsible AI is not AI that never makes mistakes. Responsible AI is AI that makes mistakes detectable, reviewable, correctable, and defensible before they become harm.

This paper defines a six-level AI incident classification model, from Level 0 blocked unsupported drafts to Level 5 reportable legal, clinical, privacy, safety, or professional events. It also proposes an incident lifecycle: Detect, Preserve, Classify, Review, Correct, Learn, and Attest.

The goal is not to turn every AI error into a crisis. The goal is to prevent organizations from ignoring small failures until they become systemic failures. A blocked unsupported draft is not a disaster. A repeated pattern of blocked unsupported drafts may be a warning. A reviewer correction is not necessarily an incident. A recurring reviewer correction around the same workflow may reveal a source-quality, prompt-design, or training problem. A near miss is not proof that AI should be abandoned. It is proof that the organization needs a disciplined method for learning from risk before risk becomes harm.

Sentinel Node, Pocono AI's local AI appliance for regulated organizations, is used in this paper as an implementation example. It is not presented as the doctrine itself. The doctrine is broader: regulated AI systems should preserve custody, evidence, review, correction, and learning loops wherever they are used.

The market implication: in healthcare and law, the most defensible AI companies will not merely generate outputs. They will control workflow, preserve evidence, reduce professional risk, and help organizations learn from failure.

1. The Governance Gap in Regulated AI

Most AI adoption conversations begin with capability. Can the system summarize a chart? Can it draft a denial appeal? Can it search case files? Can it produce a contract memo? Can it save time? These questions matter, but they are incomplete. In regulated settings, the more important questions are governance questions.

What happens when the AI is wrong? What happens when it is almost wrong? What happens when it retrieves the wrong source, omits the controlling policy, confuses one client matter with another, or generates a confident statement that lacks evidentiary support? What happens when a professional edits the output? What happens when a reviewer approves a weak draft? What happens when a payer, court, client, patient, insurer, regulator, or opposing counsel challenges the result?

Most AI systems are evaluated at the point of output. Regulated organizations need to evaluate the full chain of custody. A trustworthy AI workflow is not merely a model producing text. It is a governed process involving data custody, retrieval, source ranking, output generation, human review, correction, approval, export, audit, and post-event learning.

When those elements are missing, organizations are left with a dangerous gap. They may know that an AI output was used, but not know exactly how it was produced, what evidence supported it, who reviewed it, what was corrected, whether the output was exported, or why the system failed.

Frameworks such as the NIST AI Risk Management Framework and the NIST Generative AI Profile are important because they move AI governance away from vague principles and toward risk identification, measurement, management, monitoring, and governance. But frameworks still require operational translation. Small medical practices and small law firms need a practical doctrine that tells them what to do when AI is wrong or almost wrong.

That doctrine must be usable by real professionals under time pressure. It cannot require a Fortune 500 governance department. It must be simple enough for a clinic administrator, practice manager, attorney, paralegal supervisor, or compliance lead to understand, but rigorous enough to be defensible under scrutiny.

2. Why Near Misses Matter

A near miss is not a harmless event. It is a warning that the system came close to failure. The fact that harm did not occur does not mean the event lacked significance. In safety-conscious industries, near misses are valuable because they are cheaper, less harmful, and more instructive than adverse events.

In healthcare, a near miss may be a medication error caught before administration, a wrong-patient chart note identified before signing, or a dangerous assumption corrected during review. In legal work, a near miss may be a fabricated case citation caught before filing, an inaccurate clause summary corrected before client delivery, or a privileged document nearly exported to the wrong context.

AI near misses are especially important because AI errors can look polished. A human reviewer may see a fluent paragraph, a confident tone, or a plausible citation and assume the system is more grounded than it actually is. The danger is not only hallucination. The danger is normalized trust without preserved evidence.

Near misses should therefore be captured as operational intelligence. They can reveal weak retrieval configuration, outdated internal documents, missing source controls, inadequate reviewer training, unsafe prompts, overbroad workflow design, insufficient export controls, unsupported claims, or recurring gaps in the organization's knowledge base.

A regulated organization that fails to learn from near misses is discarding its earliest warning system. It is waiting for the system to fail externally before improving the system internally.

The near-miss doctrine changes the posture. It treats blocked drafts, reviewer corrections, retrieval mismatches, source conflicts, stale documents, and workflow boundary violations as signals. Some signals require only ordinary logging. Others require formal review. Repeated patterns require corrective action.

3. Definitions: Error, Hallucination, Near Miss, Incident, Harm

Regulated AI governance requires precise language. Without a shared vocabulary, organizations either underreact or overreact. They may dismiss serious events as harmless glitches, or treat ordinary draft corrections as crises. A useful doctrine must define terms clearly.

An error is any incorrect, incomplete, unsupported, misleading, improperly scoped, or poorly grounded output or action. Errors may arise from the model, the retrieval system, the source corpus, the user prompt, the workflow design, or the human reviewer.

A hallucination is a generated statement that appears authoritative but is not supported by the relevant evidence. In legal work, this may appear as a fabricated case, quotation, statute, rule, or holding. In healthcare, it may appear as a fabricated clinical fact, guideline, contraindication, coverage rule, or patient-specific conclusion.

A near miss is an unsafe AI event that could have produced harm, exposure, professional error, or loss of trust, but did not because it was caught, blocked, corrected, or contained before external use.

An incident is an AI-related event requiring formal review because it affected, or could reasonably have affected, a patient, client, legal matter, payer submission, court filing, compliance obligation, privacy boundary, or professional decision.

Harm occurs when an AI-related failure contributes to patient injury, client injury, legal prejudice, confidentiality breach, professional misconduct, regulatory exposure, financial loss, operational disruption, or loss of trust.

These definitions are deliberately practical. They are not intended to replace legal, medical, regulatory, or insurer definitions. They are intended to help organizations sort events quickly and consistently.

4. The Six-Level AI Incident Classification Model

The proposed AI incident model has six levels. The levels are designed to help organizations distinguish routine blocked outputs from serious incidents while still preserving enough signal to identify patterns over time.

Level 0 is a blocked unsupported draft. The system detects an unsupported, uncited, out-of-scope, or policy-violating output and prevents it from being treated as final. This is evidence that the governance layer worked. It should be logged automatically, but it does not require formal escalation unless it repeats.

Level 1 is reviewer correction. A professional reviewer identifies and corrects a weak, incomplete, or inaccurate AI draft before external use. The correction should be captured as a reviewer note or pattern marker. Over time, repeated Level 1 corrections may indicate source, prompt, or training problems.

Level 2 is a near miss. The output could have materially affected a patient, client, payer submission, legal filing, or professional decision if it had not been caught. This requires formal near-miss review.

Level 3 is an operational incident. The failure reveals a systemic weakness in the workflow, retrieval corpus, access control, reviewer process, source freshness, or AI configuration. This requires root-cause review and corrective action.

Level 4 is an externalized incident. An incorrect, unsupported, or improperly reviewed AI-assisted output reaches an external party. This requires escalation to leadership, compliance, counsel, or the responsible professional authority.

Level 5 is a reportable or high-severity event. An AI-related failure may involve patient harm, client harm, professional misconduct, privacy breach, malpractice exposure, court sanctions, regulatory reporting, or severe reputational damage. This requires formal incident response under the organization's legal, compliance, clinical, professional, and insurance processes.

Level	Name	Trigger	Minimum response
0	Blocked unsupported draft	Unsupported, uncited, or out-of-scope output is blocked before use	Automatic log; trend only if repeated
1	Reviewer correction	Professional corrects a weak or incomplete draft before external use	Reviewer note; pattern tracking
2	Near miss	Output could have materially affected patient/client/matter if not caught	Formal near-miss review
3	Operational incident	Systemic weakness in retrieval, corpus, access, workflow, review, or export controls	Root-cause review and corrective action
4	Externalized incident	AI-assisted output reaches payer, court, client, patient, or third party with defect	Escalate to leadership/compliance/counsel
5	Reportable/high severity	Potential harm, breach, sanctions, malpractice, regulatory or severe professional event	Formal incident response under governing procedures

5. The AI Incident Lifecycle

The AI near-miss doctrine requires a lifecycle, not merely a log. A log records events. A lifecycle turns events into learning, correction, and evidence.

The lifecycle begins with detection. The organization must detect unsupported claims, source gaps, retrieval failures, reviewer corrections, export attempts, unusual prompt patterns, and workflow deviations. Detection should occur

before external use whenever possible.

The second step is preservation. The organization must preserve the evidence needed to reconstruct the event. This includes the user prompt, retrieved source documents, ranking order, generated draft, citations, reviewer identity, reviewer edits, approval status, export status, timestamp, and relevant configuration information where appropriate.

The third step is classification. The event should be classified according to severity. Not every AI error is an incident. Not every incident is reportable. But every recurring pattern deserves attention.

The fourth step is review. The right professional must review the event. Healthcare events may require clinical, administrative, compliance, privacy, or payer-workflow review. Legal events may require attorney review, supervision review, privilege analysis, ethics review, or client/court communication assessment.

The fifth step is correction. Corrective action may include updating source documents, removing stale policies, revising templates, changing reviewer instructions, limiting workflows, strengthening access boundaries, or improving retrieval configuration.

The sixth step is learning. The organization should identify patterns over time. Repeated Level 0 and Level 1 events may indicate weak instructions, bad source hygiene, poor training, or unsafe workflow design.

The seventh step is attestation. The organization should be able to show evidence of governance. Attestation does not mean claiming perfection. It means showing that AI use is controlled, reviewed, documented, and improved.

Lifecycle: Detect -> Preserve -> Classify -> Review -> Correct -> Learn -> Attest. The purpose is not blame. The purpose is evidence-preserving learning.

6. Root-Cause Categories for AI Near Misses

AI failures are rarely caused by one thing. A useful incident doctrine must look beyond the model. If every error is blamed on “the AI,” the organization learns nothing. The event must be traced to the workflow component most likely to have caused or permitted the failure.

Retrieval failure occurs when the system fails to retrieve the right source, retrieves irrelevant sources, ranks weak sources too highly, or misses controlling evidence. In regulated work, retrieval failure can be more dangerous than generation failure because the model may draft fluently from the wrong evidence.

Source corpus failure occurs when the local knowledge base contains stale, duplicate, incomplete, contradictory, poorly labeled, or unauthorized documents. A local system is only as trustworthy as the curated corpus it is allowed to search.

Prompt scope failure occurs when the user asks the system to perform work outside the approved workflow, outside the user’s role, or beyond the system’s evidence base. Many AI incidents begin as ordinary convenience requests that quietly cross a professional boundary.

Generation failure occurs when the model produces unsupported synthesis, fabricated details, overconfident language, or conclusions not grounded in the retrieved evidence. This is the failure mode most people associate with hallucination, but it is only one category.

Review failure occurs when the human reviewer accepts the output without adequate verification. Human-in-the-loop governance is not magic. It requires role clarity, reviewer accountability, time to review, and tools that make verification practical.



Export-control failure occurs when a draft leaves the system before required review, approval, source support, or role authorization. In regulated settings, the boundary between draft and final work product must be visible and enforced.

Governance failure occurs when the organization lacks a policy, training structure, escalation path, documentation process, or recurring review cycle for AI events.

Root-cause category	Diagnostic question
Retrieval failure	Did the system find and rank the correct evidence?
Source corpus failure	Was the local corpus current, clean, labeled, and authorized?
Prompt scope failure	Did the user request something outside the approved workflow?
Generation failure	Did the model synthesize beyond the evidence?
Review failure	Did the reviewer have the time, role, and tools to verify?
Export-control failure	Did a non-final draft cross an external boundary?
Governance failure	Was there a missing policy, escalation path, or review cadence?

7. Healthcare Application

Healthcare organizations already operate in environments where documentation, patient safety, privacy, billing, payer policy, clinical review, and administrative overload intersect. AI can help reduce burden, but it can also create new failure modes.

A prior authorization draft may cite the wrong criterion. A denial appeal may omit a required clinical detail. A chart summary may conflate two dates. A patient message may sound final when it should be reviewed. A workflow may retrieve stale payer policy language. A staff member may trust a draft because it sounds polished.

The AI near-miss doctrine gives healthcare organizations a way to manage these risks without abandoning useful automation. It recognizes that small practices do not have unlimited staff, yet they still require evidence-preserving governance.

A healthcare AI workflow should answer: Was the output based on approved local sources? Were the sources current? Was patient data kept within the approved environment? Was the draft marked as non-final? Did a qualified human review it? Were corrections captured? Was the final output exported only after approval? If something was wrong, can the organization reconstruct the event?

For independent practices, this structure matters because they often lack the compliance staff of major hospital systems. A small clinic cannot afford an invisible AI failure. It needs simple, practical, evidence-preserving governance that fits real workflows.

The near-miss doctrine also reduces blame-oriented thinking. A flawed AI draft should not automatically be treated as individual negligence. It should be examined as a system signal: Was the source stale? Was the payer policy missing? Was the prompt too broad? Was the reviewer undertrained? Was the output label unclear? Was the workflow designed to make review realistic?

8. Legal Application

Law firms face a different but equally serious risk profile. The legal profession depends on competence, confidentiality, supervision, candor, loyalty, privilege, and truthfulness to tribunals. Generative AI can assist legal work, but it can also produce fabricated citations, false quotations, misleading summaries, privilege risks, and overconfident analysis.

Legal AI governance should therefore focus not only on productivity, but also on defensibility. A law-firm AI workflow should answer: Was client information kept inside the approved environment? Was the matter workspace isolated? Were retrieved documents actually from the correct client or matter? Were citations verified? Was the output treated as a draft? Did an attorney review the work? Were unsupported claims blocked or flagged? If challenged, can the firm show what happened?

The near-miss doctrine is especially important for law firms because many legal AI failures are preventable before filing, disclosure, or client delivery. A fabricated case caught before filing is not merely an error avoided. It is evidence that the governance system worked. A fabricated case filed with a court is evidence that the governance system failed.

Law firms should treat AI near misses as supervision intelligence. If associates, paralegals, or support staff repeatedly correct the same type of AI output, the firm should review whether the tool is retrieving the right authorities, whether staff understand citation verification, whether the workflow is limited to internal drafting, and whether the final work product approval process is clear.

The point is not to make AI taboo. The point is to keep legal judgment where it belongs: with lawyers, supported by systems that preserve evidence rather than obscure it.

9. Why Local Custody Matters

Incident response depends on evidence. If the organization cannot preserve the prompt, source set, output, reviewer action, and export status, it may not be able to reconstruct the event later. This is why local custody matters.

Cloud tools may provide convenience, but regulated organizations need to ask whether they retain enough control over client or patient data, source documents, prompt history, retrieved evidence, generated drafts, reviewer decisions, audit logs, export events, and post-event review records.

The question is not only “Where did the model run?” The question is “Can the organization prove what happened?” A governed AI system should not depend on vague trust in a vendor’s interface. It should preserve operational evidence in a form the organization can control.

Local custody also helps align incentives. When data, retrieval, inference, review, and audit are controlled locally, the organization can define what evidence is approved, who may query it, what may be exported, and how review is documented. This does not eliminate risk, but it makes risk visible.

For healthcare practices, local custody supports data-minimization and privacy-by-design principles. For law firms, local custody supports confidentiality, privilege-preservation, and matter separation. For investors, local custody supports a clearer moat: the value is not simply text generation, but controlled workflow infrastructure for regulated environments.

10. Sentinel Node as an Implementation Example

Sentinel Node is an example of how the AI near-miss doctrine can be implemented in regulated settings. The architecture is based on local custody, local retrieval, local inference, citation tracing, audit logging, and human professional review.

The goal is not to replace physicians, nurses, attorneys, paralegals, or professional judgment. The goal is to create a controlled drafting and retrieval environment where AI outputs remain reviewable, source-linked, and non-final until a responsible human approves them.

In a Sentinel Node-style workflow, the user initiates an approved workflow, the system retrieves from local approved sources, the model drafts from the retrieved evidence, the draft is marked as non-final, claims are linked to source material where possible, unsupported claims are flagged or blocked, a named reviewer approves, edits, rejects, or escalates, and the system preserves an audit envelope.

The important principle is not the brand name. The important principle is that regulated AI should create evidence as it works. The system should not merely produce a useful paragraph. It should preserve the context needed to defend, correct, and improve that paragraph.

This is why the near-miss doctrine fits Sentinel Node's core posture. A local appliance that performs retrieval, drafting, review, and audit in one governed workflow can support incident classification in a way that generic chat interfaces often cannot.

11. Corrective and Preventive Action

Incident review is incomplete unless it leads to correction. The AI near-miss doctrine therefore borrows the spirit of corrective and preventive action: identify what happened, determine why it happened, correct the immediate issue, and reduce the likelihood of recurrence.

Corrective action may be technical. The organization may remove outdated documents, improve chunking, adjust retrieval ranking, strengthen citation rules, separate workspaces, or disable unsafe workflows.

Corrective action may be operational. The organization may revise templates, clarify reviewer roles, add export gates, require second review for high-risk outputs, or train staff on the difference between draft assistance and professional judgment.

Corrective action may be governance-oriented. The organization may create a monthly near-miss review meeting, add a severity classification process, document escalation criteria, or require periodic corpus hygiene checks.

An organization should not judge AI governance by the absence of errors. It should judge AI governance by whether errors are caught early, reviewed honestly, corrected systematically, and prevented from recurring.

12. What Buyers Should Ask AI Vendors

Healthcare and legal buyers should ask vendors questions that go beyond demos. A demo answers whether the system can produce something impressive. Governance questions answer whether the system can be trusted inside regulated work.

Buyers should ask whether the system preserves prompts, retrieved sources, generated drafts, reviewer actions, and export events. They should ask whether unsupported claims can be flagged or blocked. They should ask whether outputs are clearly marked as draft or non-final.

Buyers should ask whether a human reviewer can be identified, whether patients, clients, matters, roles, and workflows can be separated, whether stale documents can be detected and removed, whether near misses can be reviewed over time, and whether the system supports internal audit without unnecessary third-party exposure.

The most important question may be the simplest: What happens when the AI is wrong? A vendor that cannot answer that question with a concrete incident-response model is not ready for regulated deployment.

Procurement teams should treat failure governance as a core requirement, not an afterthought. If a system cannot preserve evidence of failure, it cannot support disciplined learning from failure.

- Does the system preserve prompts, source sets, drafts, reviewer decisions, and export records?
- Can the system distinguish draft output from approved professional work product?
- Can the vendor explain the difference between a blocked output, reviewer correction, near miss, incident, and reportable event?
- Can the organization review repeated weak signals over time?
- Can local data custody, privilege, privacy, and matter/patient separation be demonstrated?

13. What Investors Should Notice

Investors evaluating regulated AI companies should look beyond demos. A demo shows capability. Governance shows durability.

The strongest AI companies in healthcare and law will not merely generate outputs. They will control workflow, preserve evidence, reduce professional risk, support compliance, and help organizations learn from failure.

The AI near-miss doctrine is therefore not only a safety framework. It is a market thesis. As AI becomes easier to build, defensibility will move toward custody, workflow integration, auditability, domain-specific governance, professional review, and trust infrastructure.

In regulated sectors, the winners will not be the companies that promise error-free AI. The winners will be the companies that make AI errors detectable, reviewable, correctable, and defensible.

This is especially important in the small-practice and small-firm market. These organizations need enterprise-grade governance without enterprise-scale teams. A local appliance model can be positioned as a way to deliver governance infrastructure to organizations that would otherwise be priced out of mature AI risk management.

14. Implementation Roadmap

The near-miss doctrine can be adopted in stages. Organizations should not wait for perfection before creating basic evidence-preserving controls.

Phase 1 is policy definition. The organization defines approved use cases, prohibited use cases, reviewer roles, draft labels, export rules, and incident severity levels.

Phase 2 is workflow instrumentation. The organization ensures that prompts, retrieval sets, drafts, reviewers, edits, approvals, and exports are captured for AI-assisted work products.

Phase 3 is classification. The organization begins classifying AI events from Level 0 to Level 5. The goal is consistency, not punishment.

Phase 4 is review cadence. The organization reviews Level 2 and above events promptly and reviews repeated Level 0 and Level 1 patterns periodically.

Phase 5 is corrective action. The organization connects incident review to source updates, workflow changes, reviewer training, access control, and retrieval tuning.

Phase 6 is attestation. The organization produces evidence that AI use is governed, reviewed, and improved. This evidence may be useful to leadership, auditors, insurers, investors, regulators, and professional stakeholders.

15. Conclusion

Regulated organizations do not need magical AI. They need governed AI. They need systems that respect professional responsibility, preserve sensitive information, support human judgment, and create evidence when things go wrong.

The AI near-miss doctrine gives healthcare practices and law firms a practical way to think about AI failure before it becomes patient harm, client harm, sanctions, privacy exposure, malpractice exposure, or regulatory failure.



The doctrine can be summarized in one sentence: Responsible AI is not AI that never makes mistakes. Responsible AI is AI that makes mistakes detectable, reviewable, correctable, and defensible before they become harm.

That is the standard regulated AI should meet. It is also the standard buyers, investors, clinicians, lawyers, nurses, patient advocates, and client advocates should demand.



Appendix A. Incident Intake Worksheet

The following worksheet can be adapted for a clinic, law firm, or regulated professional organization. It is intentionally simple. The goal is to capture enough information to support review without creating an unusable reporting burden.

Field	What to capture
Event date/time	When the AI event occurred or was discovered
Workflow	Prior authorization, denial appeal, legal research, contract review, document summary, intake, etc.
User/requester	Role and identity of the person who initiated the AI workflow
Reviewer	Named reviewer or responsible professional
Severity level	Level 0 through Level 5
Patient/client/matter boundary	Whether the correct boundary was preserved
Prompt/request	User request or workflow input
Sources retrieved	Documents, chunks, policies, cases, notes, or files used
Draft/output	Generated content or summary of the output
What was wrong	Unsupported claim, wrong source, stale policy, hallucination, missing fact, privilege risk, etc.
How discovered	Blocked by system, corrected by reviewer, identified after export, third-party challenge, audit review
Immediate action	Blocked, corrected, escalated, withdrawn, replaced, disclosed, investigated
Root-cause category	Retrieval, corpus, prompt, generation, review, export control, governance
Corrective action	Source update, training, workflow change, access control, retrieval tuning, policy update
Closure evidence	Who closed the event, when, and why

Appendix B. AI Near-Miss Review Checklist

A Level 2 near miss should be reviewed quickly enough that evidence is still fresh and the professionals involved can reconstruct the workflow. The following checklist is designed for practical use.

- Confirm whether the output remained internal or crossed an external boundary.
- Confirm whether the correct patient, client, matter, payer, policy, or file set was used.
- Review the retrieved evidence and determine whether the system found the correct sources.
- Review whether the generated draft stayed within the evidence.
- Identify whether the human reviewer had adequate context and authority.
- Determine whether the event was isolated or part of a recurring pattern.
- Assign a root-cause category and severity level.
- Record corrective action and assign an owner.
- Set a review date for confirming that the corrective action worked.
- Determine whether leadership, counsel, compliance, insurer, or professional board consultation is needed.

Near-miss review should be non-punitive by default. The purpose is to discover system weaknesses early enough to prevent externalized harm.

Appendix C. Governance Controls Matrix

Control area	Healthcare concern	Legal concern	Governance response
Local data custody	ePHI exposure, payer documents, chart data	Confidentiality, privilege, client files	Keep sensitive sources inside controlled environment where feasible
Source approval	Stale policy, wrong chart, wrong payer criteria	Wrong authority, wrong matter, old draft	Approved corpus, labeling, freshness checks
Citation binding	Unsupported clinical/payer claim	Unsupported legal proposition	Citation gates and source-linked drafting
Draft labeling	Patient-facing or payer-facing content treated as final	Client/court-facing content treated as final	Visible non-final status until reviewer approval
Named human review	Unclear clinical accountability	Unclear attorney supervision	Named reviewer and review status
Audit envelope	Cannot reconstruct AI-assisted administrative action	Cannot reconstruct AI-assisted legal work	Prompt, retrieval, output, reviewer, export, and timestamp evidence
Incident classification	No distinction between blocked draft and patient-safety event	No distinction between correction and sanctions-risk event	Level 0-5 model with escalation
Corrective action	Repeated stale policy or missing documentation errors	Repeated citation or matter-mixing errors	RCA and CAPA-style source/workflow/training updates

Appendix D. Website Landing Page Blueprint

The public website version should be shorter than this white paper and structured for buyer conversion.
Recommended URL: /ai-near-miss-doctrine.html.

Hero

AI Near-Miss Governance for Healthcare and Law. Sentinel Node helps regulated organizations detect, preserve, review, and learn from AI errors before they become patient harm, malpractice exposure, privilege failure, or compliance breakdown.

Recommended sections

- 1 What is an AI near miss?
- 2 The six-level AI incident classification model.
- 3 The incident lifecycle: Detect -> Preserve -> Classify -> Review -> Correct -> Learn -> Attest.
- 4 Why local custody matters.
- 5 Sentinel Node as an implementation example.
- 6 Download the full white paper.
- 7 Request a governance briefing.

Appendix E. Scenario Library for Tabletop Exercises

A doctrine becomes real when teams practice it. The following scenarios are designed for short tabletop exercises with clinical, legal, administrative, IT, and compliance participants. Each scenario should be discussed with three questions: what happened, what evidence must be preserved, and what corrective action would prevent recurrence?

Scenario	Likely level	Review focus
A denial appeal draft cites a payer criterion that was present in an outdated policy but absent from the current policy. The reviewer catches the issue before submission.	Level 2	Corpus freshness, source ranking, stale policy labeling
A law-firm memo summarizes the wrong version of a contract because both executed and draft versions were indexed with similar names. The attorney corrects the memo before sending it to the client.	Level 2 or Level 3	Matter hygiene, document status labeling, reviewer verification
A patient-facing message is drafted in a tone that appears final even though the physician has not reviewed it. The message is not sent because export requires approval.	Level 0 or Level 1	Draft labeling, export gates, reviewer workflow
A paralegal asks for case law across an entire drive instead of the approved matter workspace. The system retrieves unrelated material from another matter and flags the boundary mismatch.	Level 0 or Level 3	Access boundaries, matter separation, prompt-scope controls
A staff member repeatedly overrides warnings because the AI draft usually saves time. No external harm has occurred, but audit review shows repeated override patterns.	Level 3	Training, alert design, escalation policy, accountability
An AI-assisted filing containing an unsupported citation is filed with a court. The issue is later identified by opposing counsel.	Level 4 or Level 5	Externalized legal risk, court duty, client communication, counsel review
A patient authorization workflow generates a letter using correct diagnosis details but omits prior failed conservative treatment, leading to a weaker submission. The omission is corrected before submission.	Level 1 or Level 2	Workflow checklist, required data fields, retrieval completeness
An internal AI summary includes information from the wrong patient because two PDFs were misfiled before ingestion. The output is blocked during review.	Level 2 or Level 3	Document intake, identity verification, source corpus controls

These scenarios should be used to train the organization away from blame-first reactions and toward evidence-first reconstruction. The goal is not to ask who made the AI wrong. The goal is to determine how the workflow allowed the weakness to appear, how it was caught, and how the system should be changed.

Recommended tabletop cadence

- Quarterly for active deployments in healthcare or legal workflows.
- After any Level 3 or higher event.
- Before expanding AI use into a new workflow, practice area, payer category, or document corpus.
- Whenever new reviewer roles or new staff are added to the workflow.

Appendix F. Minimal AI Governance Policy Template

The following sample policy language can be adapted by a professional organization with counsel and compliance review. It is intentionally written as a governance skeleton rather than a final legal policy.

1. Purpose

The purpose of this policy is to ensure that AI-assisted work within the organization remains draft-only until reviewed by an authorized human professional, that sensitive data is handled within approved systems, and that AI-related errors, near misses, and incidents are captured and reviewed according to severity.

2. Approved use

AI tools may be used only for approved workflows and approved source sets. Examples may include internal document retrieval, draft generation, administrative support, denial appeal preparation, contract review support, or matter-specific research support, depending on organizational approval. AI output must not be treated as final professional advice, clinical judgment, legal judgment, or external communication without required review.

3. Human review

All AI-assisted outputs intended for external use, professional reliance, payer submission, patient communication, client communication, court submission, or formal internal decision support must be reviewed by an authorized professional or delegated reviewer under the organization's supervision model.

4. Source verification

Reviewers are responsible for verifying that material claims are supported by approved sources. Unsupported claims must be corrected, rejected, or escalated. When the system provides citations or source links, reviewers must treat them as aids to verification, not substitutes for professional judgment.

5. Incident classification

AI-related events will be classified using the organization's severity model. Level 0 and Level 1 events may be logged and trended. Level 2 events require near-miss review. Level 3 events require root-cause review. Level 4 and Level 5 events require escalation according to legal, clinical, privacy, compliance, and insurance obligations.

6. Evidence preservation

For AI-assisted workflows, the organization should preserve sufficient evidence to reconstruct the event, including the prompt or workflow request, retrieved sources, draft output, reviewer action, approval status, export event, and timestamp where feasible.

7. Non-punitive learning

Near-miss review should be conducted as a system-learning process. Deliberate misuse, concealment, unauthorized disclosure, or reckless disregard for review requirements may be subject to ordinary disciplinary or professional processes, but routine reporting of near misses should be encouraged.



Appendix G. Metrics for Near-Miss Governance

Metrics should not be used to create a false sense of precision. They should be used to detect patterns, allocate attention, and prove that the organization is learning.

Metric	Why it matters	Possible interpretation
Blocked unsupported drafts per workflow	Shows how often the system prevents weak outputs from proceeding	High rate may indicate unsafe prompts, poor source coverage, or overly ambitious workflow scope
Reviewer corrections by category	Shows where humans repeatedly fix AI outputs	Recurring corrections may justify source updates, template changes, or retraining
Near misses per month	Shows serious pre-harm signals	Increase may reflect more reporting or real workflow weakness; interpret carefully
Stale-source events	Measures corpus hygiene	High rate indicates need for document lifecycle management
Wrong-boundary events	Tracks patient/client/matter separation risk	Any recurrence should receive immediate attention
Externalized incidents	Tracks failures that crossed organizational boundaries	Requires escalation and executive review
Time to close corrective action	Shows whether learning becomes operational change	Long delays suggest governance bottlenecks
Repeat event rate	Shows whether corrective action worked	Repeat events indicate unresolved root cause

Organizations should resist the temptation to hide near misses in order to make metrics look better. A mature AI governance culture should initially expect reporting to rise as detection improves. The relevant question is not whether near misses exist. The relevant question is whether the organization knows about them, reviews them, and reduces recurrence over time.



Appendix H. 30-60-90 Day Deployment Roadmap

A small practice or law firm can adopt the near-miss doctrine in ninety days without creating a large governance bureaucracy.

Period	Operational goal	Deliverables
Days 1-30	Define boundaries	Approved use cases; prohibited uses; reviewer roles; draft labels; Level 0-5 severity definitions; initial source-corpus inventory
Days 31-60	Instrument review	Audit envelope requirements; reviewer correction capture; export gates; incident intake worksheet; near-miss escalation path
Days 61-90	Learn and attest	First monthly review; corrective action tracker; stale-source cleanup; buyer/insurer/auditor attestation summary; training refresh

The roadmap should begin with a narrow pilot. A clinic might start with prior authorization or denial appeals. A law firm might start with internal file retrieval or paralegal research support. The first pilot should be valuable enough to matter, but narrow enough that incidents can be reviewed and sources can be controlled.

Start narrow. Preserve evidence. Review near misses. Improve the workflow. Then expand.



Appendix I. RACI Model for AI Incident Governance

A simple RACI model helps prevent governance ambiguity. RACI means Responsible, Accountable, Consulted, and Informed.

Activity	Responsible	Accountable	Consulted	Informed
Approve AI workflow	Operations/IT lead	Practice owner or managing partner	Compliance, counsel, clinical/legal lead	Staff users
Review AI draft	Assigned reviewer	Licensed professional or supervising attorney/clinician	Paralegal/nurse/admin as appropriate	Requester
Classify near miss	Reviewer or compliance lead	Designated governance owner	IT, counsel, clinical/legal lead	Leadership
Conduct root-cause review	Governance owner	Practice owner/managing partner	IT, compliance, reviewer, source owner	Affected team
Approve corrective action	Governance owner	Leadership	Counsel/compliance/clinical/legal lead	Staff users
External escalation	Leadership/counsel/compliance	Practice owner/managing partner	Insurer, regulator, board, client/patient counsel as appropriate	Need-to-know only

The accountable party should never be the AI system. Accountability remains with the professional organization and its designated human roles. The system can preserve evidence, enforce gates, and support review, but it cannot carry professional responsibility.

References

[1] National Institute of Standards and Technology. AI Risk Management Framework and NIST AI 600-1 Generative AI Profile. NIST describes the Generative AI Profile as a companion resource for identifying and managing unique risks posed by generative AI.

[2] Agency for Healthcare Research and Quality, Patient Safety Network. Adverse Events, Near Misses, and Errors. AHRQ PSNet describes adverse events, near misses, and errors as patient-safety concepts and emphasizes learning from unsafe conditions.

[3] Agency for Healthcare Research and Quality, Patient Safety Network. Root Cause Analysis. AHRQ PSNet describes RCA as a structured method for analyzing serious adverse events and underlying system factors.

[4] U.S. Department of Health and Human Services, Office for Civil Rights. Summary of the HIPAA Security Rule. HHS states that covered entities and business associates must implement reasonable and appropriate administrative, physical, and technical safeguards for electronic protected health information.

[5] U.S. Department of Health and Human Services, Office for Civil Rights. Guidance on Risk Analysis. HHS describes risk analysis as an ongoing process that supports understanding risks to confidentiality, integrity, and availability of ePHI.

[6] The Joint Commission. Sentinel Event Policy and Procedures. The Joint Commission defines sentinel events as patient safety events that reach a patient and result in death, severe harm, or permanent harm, with other severe cases also considered sentinel events.

[7] American Bar Association. Formal Opinion 512 and ABA reporting on generative AI tools. ABA guidance emphasizes duties of competence, confidentiality, communication, supervision, candor, and reasonable fees in lawyers' use of generative AI.

[8] Pocono AI, LLC. Public website, Sentinel Node technical specifications, governance dashboard, and governance receipt materials. Public materials describe local inference, local retrieval, citation-bound output, human-in-the-loop review, and tamper-evident local audit trails.

[9] U.S. Food and Drug Administration. Predetermined Change Control Plan guidance for AI-enabled device software functions. FDA describes mechanisms intended to support iterative improvement while maintaining reasonable assurance of safety and effectiveness for AI-enabled devices.

End of Version 1.0. Recommended next steps: convert this doctrine into a landing page, one infographic, one LinkedIn launch post, and a downloadable PDF lead magnet.